SMS AI Tutor: Preprocessing & Feature Engineering Strategy

Phase 1 report, 04-652 AI System Design, Assignment 2 (Data Preprocessing)

1. Problem, Data, and Task

Problem. Many Rwandan upper-primary pupils have no smartphone or data. An SMS tutor sends a Science and Technology question from the REB Pupil's Book, receives the pupil's text reply, and must decide whether the reply is correct. Exact string-matching rejects correct answers that are worded differently, misspelled or written in SMS shorthand. The model replaces it with a classifier that judges a reply against the official answer (SDG 4, Quality Education).

Task type. Binary classification. Y = is_correct (1 = the reply is correct for the item, 0 = it is not).

Two tables. The raw item table is what exists today: one row per REB question with its official answer, N = 965, P = 28. The Pupil's Books and Teacher's Guides contain no pupil replies, so Y does not exist in them. The modelling table (planned) has one row per (item, candidate reply). Replies are written by a language model under a fixed specification (section 2.4) and checked against a real-life set labelled by people.

2. Part A: Dataset and Problem Audit

2.1 Initial Shape

Quantity	Value
Raw item table N x P	965 rows x 28 columns (27 data columns plus item_id)
Books read	P4 Science and Technology (Pupil's Book and Teacher's Guide), P5 Science and Technology (Pupil's Book), P6 Science and Technology (Pupil's Book and Teacher's Guide), P4 English (Teacher's Guide)
Rows by class	P4: 387, P5: 149, P6: 429
Rows by subject	Science and Elementary Technology: 896, English: 69
Rows by item type	Open question 517, multiple choice 176, fill in the blank 170, true/false 91, matching 11
Rows usable for supervised work	637 have both question text and an official answer. Of these, 580 need no picture or computer, and 538 also have an answer of at most 160 characters (it must fit in one SMS reply): open 293, fill 109, multiple choice 88, true/false 46, matching 2; P4 201, P6 337
Planned modelling table	538 items x 4 candidate replies = about 2,152 rows (N), with P roughly 25 to 30 features after encoding (section 4)
Duplicates	4 repeated item_id values (P4 English rows R01 to R04 appear twice); 41 rows repeat the same class, subject, and question text

2.2 Variable Inventory

The raw table contains counts and text but no real-valued measurements, so there are no continuous columns. Continuous features (similarity scores, overlap ratios) appear only after the reply features are derived in Phase 2.

Column	Type	Role in modelling
item_id	Identifier	Excluded (grouping key only)
subject	Nominal (2)	Feature (one-hot-encoding)
class_level	Ordinal (4 < 5 < 6)	Feature (ordinal encoding)
unit_no	Nominal (unit index inside one book)	Excluded: numbers are not comparable across books
unit_title	Text	Excluded: dirty (see 2.3), high cardinality

Column	Type	Role in modelling
section	Nominal (6)	Feature (one-hot-encoding)
q_no	Discrete (position)	Excluded: position in book carries no signal
item_type	Nominal (5)	Feature (one-hot-encoding)
stem, options, word_bank, option_labels	Text / JSON list	Used only to build features (n_options, option text for letter answers)
n_options, n_blanks, n_subitems	Discrete (counts)	Feature
n_words_stem, sms_chars, sms_segments	Discrete (counts)	Feature, after collinearity screen
needs_visual, needs_computer, uses_unicode	Binary	First two used as row filters; uses_unicode as feature
sensitive_topic	Binary	Constant False in this extract (all 965 rows): zero variance, dropped unless fixed
sms_text	Text	Excluded (duplicates stem and options)
answer_key, answer_text	Text (official answer)	Not a feature; source for the label and for similarity features
answer_key_source, stem_source, review_flag	Nominal metadata	Excluded: describe how the row was extracted, so they would leak availability

2.3 Baseline missingness

Column	Missing	%	Cause
answer_key	287	29.74%	P5 has no Teacher's Guide (149 of 149 missing); P4 Pupil's Book items the guide parser did not match; P6 guide gaps
answer_text	289	29.95%	Same rows as answer_key (plus 2)
stem	42	4.35%	40 are P4 Science Teacher's Guide answers for units whose Pupil's Book questions were not parsed (that book returned 8 of 12 units); 2 are blank question texts in the P5 and P6 Pupil's Book parse
sms_text, stem_source	40	4.15%	Follow stem
review_flag	626	64.87%	Not missing: empty means "no problem flagged"

- Overall missing cells: 4.73% of the matrix with review_flag counted, 2.58% without it.
- Complete rows (ignoring review_flag): 635 of 965;
- Rows with at least one missing value: 330.

2.4 Target label Y

Definition (planned). This is a classification problem where Y = 1 if a reply would be accepted by a teacher marking against the Teacher's Guide answer, otherwise 0.

Construction. For each of the 538 items, a language model writes four replies from the official answer, with the model name, prompt, and seed saved in data/raw/replies.csv so the step is reproducible: one correct reply, written as a pupil would send it (paraphrase, wrong word order, typos, SMS shorthand), and three incorrect replies (a typical misconception, a partly right reply that misses the key idea, an off-topic reply). The official answer is never used verbatim as a positive, since that would make "similarity = 1" a trivial rule.

Class balance. By construction, 1 correct to 3 incorrect: 25% positive, imbalance ratio 3.0.

Mitigation:

- class_weight="balanced" in the models;
- stratified, grouped splits so every fold keeps 25%;

- report precision, recall, F1, and PR-AUC for the correct class, not accuracy;
- tune the decision threshold on the validation set, because accepting a wrong reply (false positive) is worse for a tutor than asking a pupil to retry.
- Oversampling (SMOTE) is rejected: it would interpolate between embedding vectors that do not correspond to real replies.

3. Part B: Cleaning, Imputation, and Outlier Strategy

3.1 Missingness mechanism

Column group	Evidence from the data	Mechanism	Treatment
answer_key, answer_text	Missing share by class: P4 31.5%, P5 100.0%, P6 3.7%. Cramer's V with class_level 0.71 (p < 1e-100), subject 0.42, section 0.21, item type 0.16	**MAR**: depends on observed class_level and source document. Inside P4 and P6, items whose answer has several parts are skipped by the parser (72 rows flagged count_mismatch):	Never imputed (an answer cannot be guessed). Keep the row, add key_missing indicator, exclude from supervised training. Fix the parser for multi-part answers
stem, sms_text	40 of 42 missing stems are P4 Science Teacher's Guide rows for units 3, 4, 7 and others whose Pupil's Book questions were not parsed; 2 are blank texts from the P5 and P6 Pupil's Book parse	**MAR**: depends on source document layout.	Fix the P4 Pupil's Book parse, then join on (subject, class, unit, q_no); rows still missing are excluded. Add stem_missing indicator
n_words_stem, sms_chars (zeros)	All 42 rows with missing stem hold 0	Structural, same cause as stem	Recode 0 as NaN on those rows; they are removed with the rows above
count features, binary flags	No missing values	n/a	Median/mode imputers kept in the pipeline for new books

If imputation is ever needed, median/mean will be used for the numeric counts and mode for nominal columns. KNN imputation is not used: it adds cost and no benefit here, since the only columns with large gaps hold ground truth that must not be filled in.

3.2 Outlier Rules

All numeric features are right-skewed (skewness 1.8 to 13.5), so the IQR rule is the primary rule, and the Z-score rule (which assumes a bell shape) is shown for comparison only. Fences are learned on the training split only.

Feature	Skew	Q1 / Q3	IQR fence (upper)	IQR outliers	|Z|>3	Rule used
n_words_stem	8.21	7 / 14	24.5	97 (10.1%)	18	IQR upper fence, then log1p
sms_chars	7.39	52 / 123	229.5	41 (4.2%)	11	IQR upper fence, then log1p
n_options	1.81	0 / 0	0	166 (17.2%)	0	99th percentile cap (IQR = 0)
n_blanks	2.83	0 / 0	0	169 (17.5%)	49	99th percentile cap (IQR = 0)
n_subitems	4.85	0 / 0	0	52 (5.4%)	52	99th percentile cap (IQR = 0)
sms_segments	13.5	1 / 1	1	170 (17.6%)	10	99th percentile cap (IQR = 0)

3.3 Why Cap Instead of Delete

- Dropping rows would reduce N from a small base (about 538 items) and would drop the item's other features, which are valid.
- The long items are genuine (comprehension passages, multi-part tasks). Deleting them would bias the model against exactly the items that most need an SMS-length decision.

- Winsorizing at the learned fence keeps every row, and the log1p transform shrinks the remaining skew.

3.4 Association Strategy (data-type matching matrix)

Feature (type)	Against Y (binary)	Reason
Reply-to-answer features (planned): embedding cosine similarity, token overlap, edit ratio, number match (continuous, bounded)	Point-biserial r and Mann-Whitney U; mutual information	Continuous vs binary; skewed, so a rank test backs up r
Reply features: length ratio, word count (discrete, skewed)	Point-biserial on log1p values; mutual information	Count data; log makes r more meaningful
Item counts: n_options, n_blanks, n_subitems, n_words_stem, sms_chars, sms_segments (discrete)	Spearman between features; mutual information with Y	See note on item-level features below
item_type, section, subject (nominal)	Chi-square with Cramer's V; mutual information	Nominal vs binary
class_level (ordinal)	Spearman or Kendall tau-b; chi-square	Ordered categories
needs_*, uses_unicode (binary)	Chi-square (phi coefficient)	Binary vs binary

Expected result. Because every item contributes the same four replies (one correct, three wrong), the item-level columns have close to zero marginal association with Y by construction. They are not predictors on their own. They act as modifiers (for example, the similarity needed to accept a reply differs for a true/false item and an open question), so they are kept and tested through interactions and tree models, while the reply-to-answer similarity features are expected to carry most of the signal.

3.5 Multicollinearity Protocol

Computed on the raw item table: Pearson r = 0.92 for n_words_stem and sms_chars, 0.88 for sms_chars and sms_segments, 0.86 for n_words_stem and sms_segments (all above the 0.85 threshold). VIF on standardised features: sms_chars 18.0, n_words_stem 14.0, sms_segments 5.5, others below 2.5. After the log1p transform, VIF falls to 8.6, 7.9 and 2.6, below 10, but the pairwise rule still fires. sms_segments is also a direct function of sms_chars (about ceil (chars / 153)), so the two cannot both carry independent information.

- **Step 1 (pairwise):** on the training split, compute Spearman |r| between all numeric features. For every pair with |r| >= 0.85, drop the feature with the higher mean |r| to the other features. A tie goes to the feature that can be computed from the stem alone and is not a function of another feature.
- **Step 2 (VIF):** on the remaining features, compute VIF (log1p, standardised). While any VIF > 10, drop the highest and recompute.
- **Expected outcome:** drop sms_segments (a function of sms_chars) and one of n_words_stem or sms_chars. The decision is recorded in the Phase 2 decision log with the training-split numbers, which may differ from these full-table numbers.

4. Part C: Encoding, Scaling and Transformation

Feature group	Method	Justification
class_level (4, 5, 6)	Ordinal encoding to 0, 1, 2	Real order (harder content in higher classes); no extra columns
item_type (5), section (6), subject (2)	One-hot, handle_unknown="ignore"	Unordered, low cardinality (at most 6), so no more than 13 columns. A new book with an unseen type does not crash
needs_visual, needs_computer, uses_unicode	0/1 as is	Already binary
unit_no, unit_title, q_no, item_id	Not encoded	ID-like or position-only; one-hot of 14 units would add near-unique columns
Count features after the screen	Winsorize (train fences), log1p, then StandardScaler	Skew above 1.8 is removed by log1p; StandardScaler suits logistic regression and SVM baselines, which are distance and penalty based
Similarity and overlap features (bounded 0 to 1)	Left unscaled (MinMax not needed)	Already in a fixed range; scaling a bounded feature with MinMax would only make it depend on the training maximum
Reply and answer text	Frozen multilingual sentence-embedding model, used only to compute cosine similarity	Pretrained and not fit on this data, so no train/test leakage. The 384 dimensions are not fed to the classifier directly (small N); PCA on them is used for diagnostics (scree plot)

4.1 Train / validation / test partitioning

Unit of splitting: the item, not the row. All four replies of one item go to the same partition; otherwise, the model sees the official answer's wording in training and is tested on a near copy. Items with the same normalised question text (41 repeats) are merged into one group.

Ratio: 70 / 15 / 15 on groups, about 376 / 81 / 81 items (about 1,504 / 324 / 324 rows). With only a few hundred items, a larger test share would starve training, and a smaller one gives a very noisy estimate.

Tuning: 5-fold Stratified GroupKFold inside the 70% training part, stratified on item type, grouped on item. The validation 15% is used for threshold tuning.

Stratification: the label is fixed at 25% positive within every item, so stratifying on Y is automatic; stratifying on item_type and class_level keeps open, fill, multiple-choice and true/false items in each partition.

Real-life set: separate, never used in any fit or tuning step.

Why not a random row split: it leaks answers across partitions and inflates every score.

4.2 Leakage Prevention

Risk	Control
Imputer, winsor fences, scaler, encoder see test data	All steps live in one scikit-learn Pipeline / ColumnTransformer (src/pipeline_builder.py); fit is called on X_train only, and CV re-fits it inside each fold
Same item in train and test	Group split by item and normalised question text (section 4.1)
Label inferable from how the reply was generated	Generator fields (reply type, prompt id, model name) are stored in the raw table but excluded from the features, and a check confirms no feature has |r| above 0.95 with the reply type
Official answer verbatim as a positive	Positives are always rewritten; a unit test asserts no positive equals the key
Availability metadata reveals the label	key_missing, review_flag, answer_key_source and stem_source are excluded from the features
Embedding model tuned on this data	Used frozen; no fitting step touches the data

Risk	Control
Tuning on the real-life set	Locked until the final run; touched once

5. Open Items and Risks

- Extractor fixes needed: the P4 Science Pupil's Book parse (8 of 12 units), unit titles (dotted and "Unit N" values), the constant sensitive_topic, 8 duplicated English IDs, and multi-part answers that count_mismatch skips.
- English rows (69) come from one Teacher's Guide section with question text but no answer key, so they cannot be labelled. They stay in the raw table and are outside the supervised table, which holds Science and Technology only (P4 201 items, P6 337).
- Real-life test set size is not yet fixed. A minimum of 100 labelled replies is the target for a stable estimate.

6. Reproduction

pip install -r requirements.txt, then python src/extract_items.py to build data/raw/items.csv, then python phase1_strategy/phase1_audit.py to reprint every figure used in this report.